

Wealth vs Health – Does Money Buy a Longer Life?

Abstract

This study investigates whether economic wealth (GDP) is the primary driver of life expectancy. The analysis examines the relative influence of economic, educational, and health-related factors to determine whether higher national wealth necessarily translates into longer lives.

Background & Motivation

The study investigates whether economic wealth (GDP) is the primary driver of life expectancy, motivated by disparities such as the 33-year life expectancy gap between Japan (84 years) and Sierra Leone (51 years) — two nations coexisting in the same era.

Dataset & Scope

Analysis uses the WHO Global Health Observatory dataset, covering 193 countries and 2,938 records from 2000–2015 across 22 health, economic, and social indicators.

Methodology

A structured data pipeline was applied: median imputation for missing values, IQR-based outlier detection, and feature scaling, achieving 98.7% data completeness with zero nulls in the target variable. Statistical techniques — correlation analysis, PCA, and feature selection via Recursive Feature Elimination and Random Forest — evaluated the relative influence of GDP, schooling, and immunization on life expectancy.

Key Findings

The hypothesis that GDP is the dominant factor was rejected. Schooling emerged as the strongest predictor ($r = 0.75$, +1.8 years of life expectancy per additional year of schooling), followed by immunization coverage ($r = 0.62$), while GDP ranked seventh ($r = 0.46$). A multiple regression model combining schooling, immunization, and income composition achieved $R^2 = 0.83$, with GDP contributing only a marginal 0.04 improvement.

Data Equity Insight

Developing nations disproportionately suffer from missing data, revealing a “data equity” issue with direct policy implications for health monitoring.

Deliverable

Findings were operationalized into an interactive Power BI dashboard offering country-level drill-downs, comparative radar charts, and AI-generated natural language insights for non-technical policymakers.

Conclusion & Future Scope

Education and immunization access — not wealth alone — are the strongest levers for improving population health. Future work extends toward predictive modeling (XGBoost/LSTM) and a real-time policy recommendation engine.